Tayyab Shahid

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SUMMARY

Computer Science graduate from University of Education, Lahore, with 1 year of hands-on experience building and deploying Machine Learning and Generative AI projects. Comfortable working across the full ML pipeline from data preprocessing and model training to building web apps with Streamlit. Developed an AI-powered Sign Language Recognition system as a Final Year Project. Seeking a role in AI/ML.

EXPERIENCE

AI Engineer — TechAbout Private Limited, Lahore (Onsite) JULY 2026 – Present

- Working on building and deploying Machine Learning and Generative AI solutions, including RAG-based pipelines using LangChain, ChromaDB, and Hugging Face Transformers.
- Developing computer vision and deep learning models (OpenCV, MediaPipe, CNN, LSTM) and integrating them into interactive Streamlit web applications for real-time inference.

Artificial Intelligence Intern — CodeAlpha, Remote One month

- Built and deployed AI and Computer Vision applications, including an intelligent Language Translator using NLP and sequence models.
- Implemented real-time Object Detection systems using OpenCV and deep learning, covering dataset annotation, model training, and performance evaluation.

Machine Learning & AI Intern — DevelopersHub Corporation, Remote 6 weeks

- Completed a 6-week intensive remote internship focused on core Machine Learning and Data Science workflows.
- Performed EDA, feature engineering, and model training for predictive AI pipelines while collaborating in a remote agile environment.

TECHNICAL SKILLS

Languages & Libraries: Python, NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, NLTK, OpenCV, MediaPipe
Machine Learning: Supervised & Unsupervised Learning, NLP, CNN, LSTM
Generative AI: Hugging Face, LangChain, ChromaDB, RAG, LLaMA, Prompt Engineering
Tools & Deployment: Streamlit, Git, GitHub, Jupyter Notebook, VS Code

PROJECTS

DocChat — AI PDF Question Answer System (RAG) *Python | LangChain | ChromaDB | LLaMA | Streamlit*

- Developed a Retrieval-Augmented Generation (RAG) application that enables users to upload PDF documents and ask questions using natural language.
- Integrated LangChain, ChromaDB, and LLaMA to retrieve relevant document context and generate accurate responses.
- Built an interactive Streamlit interface for seamless document upload and conversational Q&A.

Talking Hands — AI Sign Language Recognition System (FYP) *Python | MediaPipe | CNN | LSTM | Hugging Face | gTTS*

- Designed an AI-powered sign language recognition system capable of translating hand gestures into speech in both English and Urdu.
- Utilized MediaPipe Holistic to capture 387 facial, hand, and body landmarks from live webcam frames for gesture recognition.
- Trained an LSTM-based deep learning model on the WLASL dataset, achieving 72% validation accuracy, with response refinement using Hugging Face Transformers.

Credit Card Fraud Detection *Python | Scikit-learn | SMOTE | Logistic Regression*

- Built a machine learning model to detect fraudulent credit card transactions on a highly imbalanced financial dataset.
- Improved minority class representation using SMOTE before training a Logistic Regression classifier.
- Evaluated performance using Precision-Recall and ROC-AUC, with emphasis on reducing false negatives in fraud detection.

Spam Email Detection System *Python | Naive Bayes | TF-IDF | NLTK | Streamlit*

- Developed an NLP-based spam detection application using text preprocessing and machine learning techniques.
- Implemented text cleaning, tokenization, stemming, and TF-IDF feature extraction before training a Naive Bayes classifier.
- Achieved 97.16% accuracy and 99% precision, and deployed the solution as a Streamlit web application.

Real-Time Object Detection & Tracking System *Python | YOLOv8 | ByteTrack | OpenCV*

- Built a real-time object detection and tracking app using YOLOv8 and ByteTrack during the CodeAlpha Internship.
- Ran frame-by-frame detection on live webcam video via OpenCV, assigning persistent tracking IDs to objects.
- Displayed a live feed with bounding boxes, class labels, confidence scores and track IDs.

EDUCATION

Bachelor of Science in Computer Science 2022 – 2026
University of Education, Lahore

Intermediate in Computer Studies 2019 – 2021
Islamia College Railway Road, Lahore

Matriculation in Computer Studies 2016 – 2018
Islamia School Multan Road, Lahore